

Predicting Bible Translation Draft Quality from Source-Side Passage Signals

James D. Cuenod

Abstract

We study whether source-side information can predict the relative quality of machine-translation drafts for Bible translation. The setting is early-stage adaptation of NLLB-style multilingual machine translation to languages with little or no previous task-specific training data: after an initial Bible passage has been translated and used for fine-tuning, the system must recommend which additional passages are likely to produce the best drafts. Across 840 experiments covering six language directions, seven training passages, and twenty candidate passages, source-side lexical overlap is a useful but incomplete predictor. Among the main self-contained similarity baselines, NLLB-tokenized TF-IDF over NIV source passages is strongest, while dense semantic embeddings, including NLLB encoder embeddings and Harrier, underperform in this particular ranking task. A learned reranker that combines train/test similarity with intrinsic candidate-passage features substantially improves ranking accuracy. These results support a source-side recommendation model rather than reliance on semantic embeddings or universal passage priors alone.

1 Introduction

Low-resource neural machine translation remains sensitive to data scarcity, domain mismatch, rare vocabulary, and the details of subword coverage (Koehn and Knowles, 2017; Sennrich et al., 2016; Ranathunga et al., 2021). These issues are especially acute in Bible translation, where many target languages have little written data and where the first substantial translated text in the language may be produced during an ongoing translation project. Recent multilingual models such as NLLB-200 make large-scale transfer possible across many languages (NLLB Team et al., 2022), and Bible-oriented corpora such as eBible have begun to provide benchmarks for low-resource Bible translation (Akerman et al., 2023). However, practical translation teams still face a local planning problem: after a first passage has been translated and used to adapt a model, which passage should be drafted next?

This paper treats that planning problem as a ranking task. Given one translated training passage and a fixed set of candidate test passages, the goal is to rank the candidates by expected machine-translation draft quality using only source-side information available before reference evaluation. This differs from standard machine-translation quality estimation. Quality-estimation systems normally score an already produced hypothesis, often with source and target-side features and sometimes with pretrained multilingual encoders or learned metrics such as COMET (Rei et al., 2020). Here the system is asked to forecast which source passage is likely to draft well before the candidate draft has been evaluated, and in the intended use case before enough target-language data exists to train a target-side estimator.

The central hypothesis is that source-side passage relationships carry actionable signal. If a candidate passage reuses lexical, subword, named-entity, or discourse material that is well represented in the initial training passage, then a fine-tuned translation model may produce a better draft.

At the same time, the candidate passage may have intrinsic properties—length, lexical diversity, proper-noun density, self-redundancy, or distributional typicality within the Bible source text—that make it generally easier or harder to draft. The experiments below test both types of signal.

The paper makes four contributions. First, it quantifies how universal observed candidate-passage rankings are across language directions and training passages. Second, it compares sparse lexical similarity, model-token similarity, and dense semantic embedding baselines. Third, it shows that a compact learned reranker using both pairwise and intrinsic source-side features substantially improves over standalone similarity. Fourth, it frames the practical output as a recommendation problem, evaluating not only rank correlation but also whether the recommended shortlist contains genuinely high-quality drafts.

2 Related Work and Positioning

Low-resource and multilingual machine translation. Neural MT systems often improve with scale and multilingual transfer, but low-resource settings remain difficult because the relevant target-language evidence is scarce, unevenly distributed, or domain-mismatched (Koehn and Knowles, 2017; Ranathunga et al., 2021). NLLB-200 is directly relevant because it was designed to expand multilingual MT coverage and includes many low-resource directions (NLLB Team et al., 2022). The present study analyzes completed NLLB-style fine-tuning runs from live Bible translation projects, but its focus is not the fine-tuning recipe itself. Instead, it asks whether source-side signals can predict which candidate passages will draft well after a project has made an initial adaptation step.

Bible translation and domain-specific MT. The Bible is a useful but unusual domain for low-resource MT: it is highly structured, verse-aligned, semantically diverse, and rich in repeated names, formulaic expressions, and genre shifts. The eBible corpus demonstrates the importance of Bible-specific data and benchmarks for many low-resource languages (Akerman et al., 2023). This study differs from corpus-building or general model benchmarking work by focusing on the order in which additional passages should be drafted after a first adaptation step.

Evaluation metrics and quality estimation. BLEU remains a standard MT evaluation metric, but it is sensitive to tokenization and reporting conventions (Papineni et al., 2002; Post, 2018). chrF and chrF++ use character n-gram matching, with chrF++ adding word n-gram information, and are often useful for morphologically varied or tokenization-sensitive settings (Popović, 2015, 2017). We use case-normalized chrF++ as the primary observed-quality metric. This is an automatic reference-overlap metric, not a direct human adequacy judgment, so the claims in this paper are about predicting metric-observed draft quality unless otherwise stated.

COMET and related learned metrics use pretrained multilingual representations and have achieved strong correlation with human judgments in standard MT evaluation settings (Rei et al., 2020). In ultra-low-resource project starts, however, target-side data may be precisely what is missing. This motivates a source-side-only predictor that can operate before a candidate passage has a usable target-language history.

Source-side similarity and representation learning. Sparse vector-space and TF-IDF representations are classical information retrieval methods (Spärck Jones, 1972; Salton et al., 1975). Their strength in this paper is not semantic sophistication but alignment with the practical mechanism being tested: reuse of source-side lexical and model-token material from the already translated passage. NLLB-tokenized TF-IDF additionally uses the model-facing subword inventory; subword

segmentation is a standard response to rare-word and open-vocabulary problems in MT (Sennrich et al., 2016; Kudo and Richardson, 2018).

Dense multilingual embeddings offer a contrasting hypothesis: semantically similar passages should be good candidates even when they do not share surface vocabulary. We test NLLB encoder embeddings and Harrier, a recent multilingual embedding model released by Microsoft for retrieval and semantic similarity (Microsoft, 2026). These tests should be read as evaluations of the specific embedding configurations used here, not as a general rejection of dense semantic representations.

3 Task Definition

The practical problem is a ranking problem arising during Bible translation for languages with limited training data. Suppose a team has fine-tuned an NLLB-style model for a new or previously unseen language using one translated source passage and can generate drafts for several candidate passages. We want to predict which candidate drafts will be highest quality, using only source-side information available before reference evaluation.

For each language direction d , training passage t , and candidate passage c , the experiment observes a draft-quality score $q(d, t, c)$ measured by case-normalized chrF++. Candidate passages are then ranked within each (d, t) group. Rank 1 is the best observed draft. A predictor assigns a source-side score $s(d, t, c)$, and candidates are ranked by that score. The evaluation asks how closely the predicted ranking recovers the observed ranking.

This formulation intentionally avoids target-side features. It is therefore not a replacement for human checking, post-editing, or target-side quality estimation once drafts exist. It is a pre-evaluation recommendation layer for deciding where a draft is likely to be most useful next.

4 Data and Evaluation

The dataset contains 840 completed draft-quality experiments (Table 1). For each language direction d and training passage t , we rank the twenty candidate passages c by case-normalized chrF++. The source-side proxy text is NIV11. All source texts are mapped to the same canonical verse references and normalized for whitespace and common quotation/dash variants before passage-level features are computed.

Quantity	Count
Language directions	6
Training passages	7
Candidate test passages	20
Rows	840

Table 1: Experimental grid. Full language directions and passage ranges are listed in Appendix A.

For each language-direction/training-passage group, we report three ranking metrics. Mean pair-Spearman is the Spearman correlation between predicted and observed ranks, averaged over all (d, t) groups. MAE is mean absolute rank error, $|\hat{r} - r|$. Within-3 is the proportion of candidates whose predicted rank is within three positions of the observed rank. For recommendation shortlists, we additionally report Top-1 in top-3, Precision@3, Precision@5, and the mean observed rank of the recommended top five passages.

BLEU and chrF++ are strongly aligned in these experiments: across all rows, BLEU-lowercase and chrF++-lowercase have Spearman correlation 0.957, and within the 42 language-direction/training-passage ranking groups their mean Spearman correlation is 0.865. Divergence is mainly local rather than systematic: the lowest within-group correlations occur for Amharic-to-Wolaytta after Micah–Haggai training (0.642) and Swahili-to-Tonga after Micah–Haggai training (0.749), and the top-ranked passage agrees in 76.2% of groups. We use chrF++ because character n-gram matching is generally more stable than word-token BLEU for ultra-low-resource languages, scripts, and tokenization conventions, while the word n-gram component in chrF++ preserves some word-level precision information (Popović, 2015, 2017).

5 How Universal Are Passage Rankings?

The first question is whether passage rankings are universal. If all language pairs rank candidate passages similarly, a simple global prior could be enough. The results show partial universality, but not enough for recommendation (Table 2).

Vantage point	Unit correlated	Mean Spearman
Language pairs, fixed training passage	20 test ranks	0.683
Training passages, fixed language pair	20 test ranks	0.407
Language pairs, fixed test passage	7 training-passage profile	0.680
Language pairs, full rank surface	7×20 ranks	0.683

Table 2: Agreement of actual rankings from four vantage points. “Language pairs, fixed training passage” compares two language directions while holding the training passage fixed and correlating the 20 test ranks. “Training passages, fixed language pair” compares two training passages within the same language direction. “Language pairs, fixed test passage” compares the seven-rank profile of one test passage across training passages. “Full rank surface” correlates all 7×20 ranks between two language directions.

Test passages have stable tendencies across language pairs, but training-passage interactions remain important. The agreement between different training passages within a language pair is much lower (0.407), confirming that the interaction between the training data and the source side of the test data impacts the quality of drafts. Therefore, both the training data and the draft candidate text must be accounted for in any predictive model.

Each vantage point tests a different possible source of usable information. Holding the training passage fixed asks whether language directions broadly agree about which candidate passages are easy after the same starting point. Holding the language direction fixed and comparing training passages asks how much the starting passage changes the ranking. Holding the test passage fixed asks whether a candidate passage has a stable profile of dependence on the training passage across language directions. This last quantity is less directly actionable, but it helps diagnose whether train/test interactions are language-specific or partly transferable. The full rank surface summarizes the overall agreement between language directions across all observed training/test combinations.

This is confirmed by leave-one-direction-out consensus baselines (Table 3). The universal prior is much better than random, but still too weak. The train-specific consensus is much stronger, showing that transferable train/test structure exists. The apparent consensus ceiling is descriptive only, since it uses all directions including the evaluation direction.

Predictor	Spearman	MAE	Within-3
Universal test-passage prior, ignores training passage	0.523	4.25	0.543
Train-specific consensus, leave-one-direction-out	0.778	2.78	0.687
Apparent train-specific consensus ceiling	0.845	2.30	0.768
Random ranking	0.000	6.66	0.319

Table 3: Consensus and random baselines. The universal test-passage prior averages each test passage’s rank across other directions and ignores the training passage. The train-specific consensus averages each train/test pair across other directions.

A critical implication is that the strongest non-source baseline is not a deployable solution for a first project in isolation. The train-specific consensus requires historical outcomes from other language directions over the same training and candidate passages. It is useful as a benchmark for transferable structure, but a source-side model is preferable when candidate sets change or when historical outcomes are sparse.

6 Source-Side Similarity Baselines

We evaluated several source-side similarity strategies (Table 4). These strategies differ in the textual representation, the granularity of comparison, and the way verse-level similarities are aggregated into a passage-level score.

Passage-level TF-IDF is the simplest strategy. It represents the full training passage and the full candidate passage as two sparse lexical vectors and uses their cosine similarity as the score. This asks whether the two passages have similar overall vocabulary, but it does not distinguish one concentrated high-overlap region from many weaker overlaps. The representation follows the classical vector-space and term-weighting tradition in information retrieval (Spärck Jones, 1972; Salton et al., 1975).

Verse-window TF-IDF is more local. Each verse is embedded separately, and we compute the full matrix of pairwise similarities between every training verse and every candidate-test verse. The top-10% score flattens this entire train-by-test matrix and averages the strongest tenth of all verse pairs; it is not computed separately for each test verse. The q95 score is the 95th percentile of the same matrix. The target-best score is asymmetric: for each candidate-test verse, it takes the best matching training verse and then averages over candidate-test verses. This last aggregation is closest to a test-side coverage question, but it was not the strongest performer.

The NLLB-token TF-IDF rows use the same sparse TF-IDF logic, but tokenize the source text with the NLLB tokenizer before fitting the vectorizer. This tests whether overlap in model-facing subword units is more informative than overlap in ordinary words or character n-grams. Subword modeling is a standard response to rare-word and open-vocabulary problems in neural MT (Sennrich et al., 2016; Kudo and Richardson, 2018).^{1 2}

¹The NLLB-token TF-IDF features use the `facebook/nllb-200-distilled-600M` tokenizer as a model-facing subword inventory. NIV rows set the tokenizer source language to `eng_Latn`; source-specific rows set it to the direction’s source language. Tokenization removes special tokens, does not truncate verses, keeps NLLB piece case, and fits a sublinear TF-IDF vectorizer over the resulting token strings. This is a source-representation choice and should not be read as a claim about the exact checkpoint used for draft generation.

²We also explored maximal marginal relevance (MMR) as an alternative aggregation for NLLB-token TF-IDF. For each candidate-test verse, MMR greedily selects k training verses using a score of the form $\lambda \text{sim} - (1 - \lambda) \text{redundancy}$, where larger λ favors similarity and smaller λ favors diversity. We swept $k \in \{3, 5, 8, 10, 12, 15\}$, $\lambda \in [0.20, 0.99]$, and

The NIV rows use a single English source-side proxy for all directions. The source-specific rows instead compute the same kind of signal in the actual source language used for each direction. This distinction is important because a recommendation system may have either a stable pivot source, such as NIV, or a direction-specific source text. The stronger NIV results should not be interpreted as evidence that English is inherently superior as a source; they may reflect consistency of preprocessing, quality of the NIV source proxy, tokenizer behavior, or differences among the source-specific texts.

The semantic embedding rows replace sparse lexical vectors with dense multilingual representations. Harrier is included because it is a recent multilingual embedding model released by Microsoft for retrieval and semantic similarity (Microsoft, 2026), so it provides a useful check on whether a general-purpose semantic representation can recover quality-relevant passage relationships better than lexical overlap. The reported Harrier centroid score uses `microsoft/harrier-oss-v1-0.6b`, embeds verses, averages them into one vector per passage, and compares the training and candidate passage centroids. Other Harrier pairwise variants used a retrieval-style prompt for the candidate-test verse side, but the centroid row reported in Table 4 is the strongest Harrier result.

The NLLB encoder rows use mean-pooled final encoder states from `facebook/nllb-200-distilled-600M`, with full verses pooled over non-padding tokens and L2-normalized before cosine similarity. For symmetric MaxSim, each candidate-test verse is matched to its most similar training verse, each training verse is matched to its most similar candidate-test verse, and the two averages are themselves averaged. We also tested target-best variants over the candidate-test verses: mean target-best, q95 target-best, and top-10% target-best. The top-10% target-best variant is strongest for NIV NLLB encoder embeddings, while symmetric MaxSim remains strongest for the source-specific NLLB encoder embeddings by rank correlation.

Strategy	Spearman	MAE	Within-3
NIV verse NLLB-token TF-IDF, top-10% mean	0.681	3.38	0.607
NIV verse NLLB-token TF-IDF, q95	0.673	3.43	0.614
NIV verse word TF-IDF, target-best mean	0.671	3.47	0.613
NIV verse char(3,5) TF-IDF, top-10% mean	0.663	3.54	0.583
Source-specific verse NLLB-token TF-IDF, top-10% mean	0.645	3.59	0.579
NIV passage word TF-IDF	0.641	3.74	0.564
Source-specific verse char(3,5) TF-IDF, top-10% mean	0.624	3.71	0.589
Dense semantic Harrier centroid	0.561	4.06	0.544
NIV NLLB encoder embeddings, target-best top-10%	0.532	4.14	0.551
Source-specific NLLB encoder embeddings, symmetric MaxSim	0.505	4.36	0.500

Table 4: Source-side similarity baselines. NIV-based rows use the shared NIV source text for all directions. Source-specific rows use the direction’s own source text. NLLB encoder rows use mean-pooled, L2-normalized encoder embeddings with symmetric MaxSim aggregation.

Three findings matter. First, TF-IDF is a strong source-side proxy. It outperforms dense semantic baselines, which is plausible in this low-resource setting: lexical and source-token coverage is closer to what the translation model has learned than broad semantic similarity. Second, NLLB-tokenized TF-IDF is the strongest main self-contained baseline, indicating that model-token overlap

several summaries of the selected similarities. The best standalone MMR result was strongly similarity-weighted: $k = 5$, $\lambda = .99$, using the 75th percentile of the selected-verse minimum similarities, gave Spearman 0.703, MAE 3.23, and Within-3 0.619, compared with Spearman 0.681 for the NLLB-token top-10% baseline (Carbonell and Goldstein, 1998). More diversity-aware settings were competitive but weaker: $k = 8$, $\lambda = .75$ gave Spearman 0.697, and $k = 5$, $\lambda = .65$ gave 0.695. The gains did not clearly survive learned reranking under the held-out test-passage evaluation; see Section 7.

is more predictive than ordinary word or character overlap. Third, the NIV proxy is stronger than the source-specific multilingual alternatives tested so far, including source-specific NLLB semantic embeddings.

Note that these results only show that the *tested embedding configurations* were weaker than sparse overlap features for *this specific task*: predicting relative chrF++ draft quality after passage-level adaptation. It is possible that better task prompting, fine-tuned embedding models, or pairwise learned embedding features could change the comparison.

7 Learned Similarity-plus-Intrinsic Reranking

Similarity alone does not fully explain quality. We therefore trained a compact reranker to predict within-group standardized quality using both similarity features and intrinsic passage features. Intrinsic features include lexical diversity, verse length, corpus-relative entropy, proper-noun density, self-redundancy, token count, and train/test token coverage.

Intrinsic features are worth examining because the universality results show that candidate passages have stable tendencies across language directions. If some passages are generally easier or harder for the translation system, independent of the particular training passage, then a pure train/test similarity score will miss part of the signal. Intrinsic features are intended to capture this passage-level difficulty.

The intrinsic and pairwise features have direct signal (Table 5). For this descriptive analysis, quality is first standardized within each language-direction/training-passage group: we subtract that group’s mean quality and divide by its standard deviation. The table then reports the Pearson correlation between each feature and this within-group standardized quality over the full set of completed experiments. This correlation table is not itself a leave-one-out model; the leave-one-out results are reported separately in Table 6. The purpose here is to identify which source-side quantities carry direct signal.

The strongest features are coverage features. Word frequency coverage measures the proportion of candidate-test word tokens whose lowercased word form appears at least once in the training passage. It is frequency-weighted because repeated test word tokens are counted repeatedly. Distinct-word coverage is the corresponding vocabulary-level measure: the proportion of distinct lowercased word forms in the candidate-test passage that also occur in the training passage. These features directly test the practical hypothesis that drafts are better when the candidate passage reuses lexical material already represented in the translated training passage.

Proper-noun density is computed from a comprehensive proper-noun inventory derived from a tagged original language corpus aligned to the NIV11 text. The “without top names” rows repeat the calculation after excluding the ten most frequent names (Lord, David, Judah, Jerusalem, Jesus, Christ, Saul, Moses, Egypt, Israel), so the result is not driven only by the most common biblical names.

Self-redundancy is the mean off-diagonal verse similarity inside a passage: each verse is compared with every other verse in the same passage, and the average non-self similarity is used as the feature. Higher self-redundancy means the passage repeats more of its own lexical material, which may make it easier for a translation system to handle consistently.

Length and diversity features describe the candidate passage itself. Mean verse length is the average number of word tokens per verse. Token count is the total passage length. Type-token ratio is the number of distinct lowercased word forms divided by the total number of word tokens, so higher values indicate greater lexical diversity. In these data, higher type-token ratio is negatively correlated with quality, consistent with the idea that more varied passages are harder.

Entropy features compare the passage’s word distribution with the broader NIV11 distribution. Plain token entropy measures diversity within the passage alone. Cross-entropy under the whole-NIV11 distribution and KL divergence from that distribution measure how atypical the passage’s vocabulary is relative to the larger source text. The negative correlations suggest that more distributionally atypical passages tend to rank lower.

Feature	Pearson correlation
Word frequency coverage of test tokens by training passage	0.612
NIV11 proper-noun density	0.391
Distinct-word coverage of test vocabulary by training passage	0.373
Unique proper-noun density without top 10 names	0.311
Proper-noun density without top 10 names	0.307
Self-redundancy	0.211
Mean verse length	0.191
Token count	0.146
Plain token entropy	-0.086
Type-token ratio	-0.248
KL divergence from whole-NIV token distribution	-0.303
Cross-entropy under whole-NIV token distribution	-0.415

Table 5: Selected feature correlations with within-group standardized case-normalized chrF++. Positive values indicate higher predicted draft quality.

The learned reranker substantially improves over standalone TF-IDF (Table 6). These are leave-one-out style evaluations. In each case, one value of the named grouping variable is held out, the model is fit on all remaining rows, and predictions are made only for the held-out rows. The process is repeated until every row has an out-of-fold prediction. The leave-training-passage split asks whether the learned relationship generalizes to a new starting passage. The leave-test-passage split asks whether the model generalizes to a candidate passage whose outcomes were never seen in training, which guards against merely learning passage identity. The leave-direction split is closest to the intended use case: a new language direction with no observed quality outcomes.

The main learned rows in Table 6 use a compact gradient boosting model (Friedman, 2001; Pedregosa et al., 2011). A Ridge similarity-only row is included as a linear sanity check: it asks whether a simple weighted combination of similarity features is enough before adding nonlinear interactions or intrinsic passage features.³ This keeps the comparison focused on validation regime and feature set rather than on extensive model selection.

The leave-direction sim+intrinsic result nearly matches the train-specific leave-one-direction consensus baseline (Spearman 0.778, MAE 2.78). This is the strongest evidence that the recommendation model should not be a pure similarity ranker.⁴

³The gradient boosting reranker uses scikit-learn 1.4.2’s `GradientBoostingRegressor` with 200 estimators, depth-2 trees, learning rate 0.05, subsampling 0.8, and fixed random seeds; reported sim+intrinsic results are averaged over eight seeds. These hyperparameters were fixed across the comparisons reported here rather than selected by a nested validation procedure. The model predicts within-group standardized chrF++ and is evaluated by ranking the out-of-fold predictions within each language-direction/training-passage group.

⁴Adding the full MMR feature family to the gradient boosting reranker, with MMR features selected inside each training fold, did not materially improve the learned model. In the leave-direction setting, Spearman changed from 0.785 for the base source+intrinsic+coverage feature set to 0.786 with the best fold-selected MMR setting. In the stricter leave-test-passage setting, the base feature set scored 0.701, while the best MMR-augmented setting in the sweep scored 0.682. This suggests that MMR is useful as a standalone similarity aggregation, but that its signal is

Validation scheme	Feature set	Spearman	MAE	Within-3
No split	best single feature	0.671	3.47	0.613
Leave-training-passage	similarity only (Ridge)	0.674	3.47	0.608
Leave-training-passage	similarity only (GBM)	0.661	3.53	0.589
Leave-training-passage	similarity + intrinsic (GBM)	0.787	2.76	0.698
Leave-test-passage	similarity only (GBM)	0.647	3.63	0.590
Leave-test-passage	similarity + intrinsic (GBM)	0.711	3.17	0.655
Leave-direction	similarity + intrinsic (GBM)	0.775	2.77	0.682

Table 6: Learned reranking results. GBM rows use a compact gradient boosting model trained to predict within-group standardized quality, then rank candidate passages from out-of-fold predictions. The Ridge row is a linear similarity-only comparison. MAE and Within-3 are rank-error metrics.

8 Recommendation Findings

The practical output of this work is not only a complete ranking, but a short set of plausible next passages for a translator or project manager to consider. Table 7 therefore evaluates both the first recommendation and the quality of the recommended top- k set.

For each held-out language direction, the ranking method is applied to each training passage and asked to order the 20 candidate test passages. Consensus methods are trained only on the other language directions. Source-side TF-IDF methods do not use draft-quality information from the held-out direction; they rank candidates from the source texts alone. “Top-1 in top-3” is the probability that the single highest-ranked recommendation is among the three best observed drafts. “Precision@3” and “Precision@5” measure the fraction of the recommended top 3 or top 5 passages that are truly in the observed top 3 or top 5. These list-level metrics are important because a real system may present several options and allow the user to choose among them.⁵

For a new language pair, the most relevant validation scheme is leave-direction-out. For a new set of candidate passages, leave-test-passage is the more conservative estimate. Leave-training-passage performance is still useful, but it is the least conservative if the same candidate test passages recur across folds. In the leave-direction setting, the NIV NLLB-token TF-IDF top recommendation falls in the actual top three passages 85.7% of the time (Table 7). Its recommended top five has mean observed rank 5.12, much better than the random expectation of 10.50, indicating that even the broader recommendation set contains useful ordering information.

9 Critical Discussion

The most important limitation is that the target metric and the strongest source-side predictors are both partly character- or token-overlap based. chrF++ rewards character n-gram overlap with a reference translation, while TF-IDF predictors reward source-side lexical or subword overlap between the training and candidate passages. This does not make the result circular, since the

either already captured by the learned features or does not generalize cleanly to unseen candidate passages. This MMR comparison used the expanded source+intrinsic+coverage feature set built for the MMR sweep, so its 0.785 baseline is not intended to replace the compact Table 6 leave-direction row.

⁵The MMR sweep was more encouraging for shortlist quality than for the final learned reranker. The strongest standalone MMR shortlist variants reached Precision@5 of 0.748, compared with 0.671 for the NLLB-token top-10% baseline. Thus MMR may be useful when the system needs to present several plausible passages rather than identify a single best next passage.

Strategy	Spearman	Top-1 in top-3	Precision@3	Precision@5	Mean rank@5
Train-specific consensus, leave-direction	0.778	0.929	0.738	0.762	4.33
NIV NLLB-token TF-IDF, verse top-10%	0.681	0.857	0.683	0.671	5.12
NIV char(3,5) TF-IDF, verse top-10%	0.663	0.833	0.603	0.681	4.99
NIV word TF-IDF, verse top-10%	0.665	0.810	0.571	0.690	5.09
Universal passage prior, leave-direction	0.523	0.357	0.349	0.505	6.77
Random recommendation	0.000	0.150	0.150	0.250	10.50

Table 7: Recommendation quality for a new language direction. Each row ranks the 20 candidate test passages for each held-out direction/training-passage case. Mean rank@5 is the mean observed rank of the recommended top five passages.

features are computed only from source texts and the target-side metric is computed only after draft generation. It does, however, mean that the present evidence is strongest for predicting reference-overlap quality rather than all dimensions of translation usefulness. In an ultra-low-resource setting this is partly unavoidable: reliable semantic quality estimation is precisely what is least available, and character-level reference metrics are often more stable than word-token metrics across scripts and tokenization conventions.

The semantic-embedding results are also bounded by the configurations tested here. Harrier and NLLB encoder embeddings are strong general-purpose multilingual representations, but the experiments used fixed off-the-shelf embeddings rather than task-tuned encoders or a learned cross-encoder. Their underperformance therefore suggests that broad semantic similarity is not enough for this ranking task under the present conditions; it does not rule out other prompting strategies or semantic models trained specifically for this setting.

Finally, the experiments are small in the statistical sense: six language directions, seven training passages, and twenty candidate passages. The leave-direction results are the most relevant proxy for a new language direction, but they still assume that future projects resemble this passage grid and this domain. The recommendation should therefore be treated as a decision aid for draft triage, not as a replacement for translator judgment.

10 Future Work

Future work should validate the same ranking and reranking strategy across more language directions, more scripts, and more typologically varied source/target pairs. It should also test whether the source-side predictors remain useful when evaluated against human adequacy judgments or other metrics less directly tied to surface-form overlap.

Two extensions are especially important for deployment. First, the learned reranker should be tested when the candidate passage set changes, since leave-test-passage performance is the conservative estimate for genuinely new source passages. Second, semantic features should be revisited with task-adapted encoders, prompt variants, and possibly learned pairwise models that can make better use of dense representations than simple centroid or MaxSim aggregation.

11 Conclusion

Source-side signals can predict relative Bible translation draft quality well enough to support draft triage. A universal passage prior is insufficient, and dense semantic similarity underperforms TF-IDF in this setting. The strongest main self-contained similarity baseline is NLLB-tokenized verse-pair TF-IDF over the NIV source, with exploratory MMR aggregations offering additional standalone gains but not a robust learned-reranker improvement. The strongest practical direction is a learned sim+intrinsic reranker, which combines train/test source overlap, intrinsic candidate difficulty, proper-noun features, corpus-relative entropy, and token coverage.

A Language Directions and Passage Ranges

Source code	Source language	Target code	Target language
amh_Ethi	Amharic	wal_Latn	Wolaytta
eng_Latn	English	kpg_Latn	Kapingamarangi
fra_Latn	French	ktu_Latn	Kituba
spa_Latn	Spanish	gar_Latn	Garifuna
spa_Latn	Spanish	grn_Latn	Guarani
swh_Latn	Swahili	toi_Latn	Tonga

Table 8: Language Directions.

Training passage	Verses	Tokens	Mean v. tokens	TTR	PN/100
GEN 12:1–GEN 25:34	394	9340	23.7	0.128	6.6
2CH 10:1–2CH 20:37	229	6009	26.2	0.183	9.7
LUK 10:1–LUK 14:35	225	5146	22.9	0.203	1.2
MIC 1:1–HAG 2:23	299	7815	26.1	0.204	3.8
JOS 1:1–JOS 24:33	658	16292	24.8	0.108	9.1
PRO 1:1–PRO 9:18	256	3826	14.9	0.236	0.5
2CO 1:1–2CO 13:13	256	6255	24.4	0.181	1.6

Table 9: Training passage ranges and intrinsic NIV11 metadata. Tokens are word-level NIV11 tokens; PN/100 is proper-noun mentions per 100 tokens from a comprehensive NIV11-derived proper-noun inventory.

Candidate test passage	Verses	Tokens	Mean v. tokens	TTR	PN/100
EXO 16:1–EXO 24:18	248	6112	24.6	0.171	3.9
RUT 1:1–RUT 4:22	85	2391	28.1	0.220	5.3
JER 21:1–JER 33:26	361	10443	28.9	0.130	6.3
GEN 26:1–GEN 36:43	391	9058	23.2	0.135	7.5
PRO 25:1–PRO 31:31	202	3321	16.4	0.293	0.5
1TH 1:1–2TH 3:18	136	3070	22.6	0.213	2.2
MAT 8:1–MAT 9:34	68	1423	20.9	0.297	2.2
ISA 28:1–ISA 35:10	166	4306	25.9	0.256	2.2
REV 1:1–REV 3:22	71	2006	28.3	0.244	2.1
2KI 1:1–2KI 17:41	491	14798	30.1	0.108	7.2
JER 1:1–JER 20:18	494	13588	27.5	0.131	3.4
ROM 1:1–ROM 11:36	315	7276	23.1	0.147	1.6
MRK 5:1–MRK 8:38	173	3591	20.8	0.215	1.9
LEV 1:1–LEV 16:34	490	13137	26.8	0.075	2.1
1JN 1:1–3JN 1:14	132	3142	23.8	0.147	0.9
PSA 135:1–PSA 150:6	225	3572	15.9	0.226	3.7
ECC 1:1–ECC 12:14	222	5039	22.7	0.199	0.1
JHN 1:1–JHN 4:54	166	3281	19.8	0.200	4.1
ACT 1:1–ACT 12:25	438	9772	22.3	0.150	4.5
1KI 12:1–1KI 22:54	382	10674	27.9	0.123	7.5

Table 10: Candidate test passage ranges and intrinsic NIV11 metadata. Tokens are word-level NIV11 tokens; PN/100 is proper-noun mentions per 100 tokens from a comprehensive NIV11-derived proper-noun inventory.

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